

SIOUX LOOKOUT, ON, Canada

WMO: 718420

Lat: 50.121N

Lon: 91.903W

Elev: 1257

StdP: 14.04

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-27.1	-22.4	-34.8	0.8	-26.7	-30.5	1.0	-22.1	18.9	7.4	17.4	4.7	4.1	290	0.633

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	18.6	83.8	67.8	80.8	65.4	78.0	64.1	70.7	80.4	68.6	77.1	66.6	74.5	9.4	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.5	105.9	75.5	65.5	98.6	73.2	63.6	92.2	71.1	35.5	81.0	33.6	76.8	32.0	74.6	77.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.3	16.1	14.1	-35.6	90.0	5.4	3.6	-39.5	92.7	-42.7	94.8	-45.7	96.8	-49.6	99.5		
			-35.8	74.0	5.3	1.9	-39.7	75.3	-42.8	76.5	-45.8	77.5	-49.7	78.9		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	36.2	1.5	6.4	20.7	35.7	49.5	61.2	65.8	63.8	54.8	40.6	24.4	8.6
	DBStd	24.92	14.49	13.74	13.97	10.82	9.30	7.55	6.03	6.38	8.49	8.81	11.93	14.02
	HDD50	6628	1504	1222	911	440	126	8	0	1	47	317	769	1283
	HDD65	10712	1969	1642	1374	880	485	157	62	100	319	757	1219	1748
	CDD50	1599	0	0	1	9	112	343	491	427	191	25	0	0
	CDD65	212	0	0	0	0	6	42	88	62	14	0	0	0
	CDH74	1706	0	0	0	3	75	404	680	448	94	2	0	0
	CDH80	354	0	0	0	0	9	92	149	92	12	0	0	0
Wind	WSAvg	7.1	6.8	6.9	7.2	7.3	7.5	6.8	6.5	6.7	7.1	7.8	7.8	6.9
Precipitation	PrecAvg	29.1	1.5	1.1	1.6	1.8	2.6	3.9	3.7	3.6	3.3	2.5	2.0	1.5
	PrecMax	40.0	3.7	2.2	4.1	5.1	5.3	6.3	6.4	7.9	6.5	6.4	3.7	2.3
	PrecMin	19.8	0.4	0.4	0.3	0.2	0.6	1.6	1.2	0.8	0.7	0.9	0.5	0.7
	PrecStd	5.1	0.9	0.5	0.8	1.2	1.2	1.3	1.3	1.8	1.6	1.3	0.9	0.4
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	35.5	41.3	57.9	69.4	80.5	88.5	88.8	86.9	82.0	70.0	54.7	38.4
		MCWB	33.5	37.0	48.4	51.4	61.6	70.3	70.9	69.6	66.4	58.6	49.1	34.6
	2%	DB	31.6	35.5	49.5	62.3	75.6	81.9	83.7	82.4	76.7	63.8	48.1	33.4
		MCWB	30.4	32.2	41.0	47.2	57.5	64.4	68.1	67.8	65.0	55.4	43.1	31.4
	5%	DB	27.5	32.0	43.8	57.1	70.9	78.6	80.8	78.9	72.3	58.8	43.1	30.9
		MCWB	26.1	29.7	37.3	44.5	55.1	62.8	66.2	65.7	62.0	51.6	39.9	29.6
	10%	DB	22.8	27.5	38.6	52.8	66.2	75.1	77.8	75.6	68.4	54.0	39.4	28.0
		MCWB	21.5	25.3	33.7	42.0	52.9	61.2	64.6	64.0	60.3	49.1	36.4	26.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.9	37.6	49.9	53.6	65.3	72.1	73.5	72.7	69.8	61.2	49.9	35.0
		MCDB	35.4	41.1	57.8	66.4	74.7	84.0	83.5	82.4	77.4	66.0	54.0	37.5
	2%	WB	30.4	32.7	42.4	48.9	61.0	68.5	71.0	69.7	66.5	56.4	44.1	32.1
		MCDB	31.6	35.0	48.6	59.2	69.8	77.8	80.3	78.8	73.3	62.5	47.1	33.2
	5%	WB	26.2	29.7	37.7	45.7	57.5	65.8	68.9	67.8	63.9	52.7	40.3	29.6
		MCDB	27.5	32.2	42.7	55.8	67.5	73.7	77.3	76.0	70.5	57.9	42.9	30.9
	10%	WB	21.6	25.2	34.1	42.7	54.5	63.4	66.7	65.8	61.3	48.9	36.4	26.7
		MCDB	22.8	27.6	38.2	51.7	64.1	71.6	74.7	73.1	67.0	53.2	39.2	28.1
Mean Daily Temperature Range	5% DB	MDBR	17.0	19.8	20.8	20.5	20.6	19.5	18.6	18.3	16.4	12.8	11.7	14.2
		MCDBR	14.9	17.1	21.5	27.4	27.1	24.1	22.0	21.9	21.7	21.1	14.6	13.8
		MCWBR	13.9	14.7	14.9	16.5	13.6	10.6	9.8	10.3	12.0	13.7	10.8	12.4
	5% WB	MCDBR	14.9	15.3	19.7	24.8	22.0	19.8	19.2	19.0	18.7	19.3	13.9	12.9
		MCWBR	14.1	13.4	14.4	16.1	12.7	11.0	10.2	10.4	11.9	13.6	10.9	11.7
Clear-Sky Solar Irradiance	taub		0.250	0.257	0.284	0.321	0.345	0.363	0.384	0.361	0.348	0.297	0.277	0.251
	taud		2.306	2.300	2.344	2.420	2.455	2.431	2.399	2.474	2.491	2.602	2.477	2.332
	Ebn at Noon		256	284	295	294	289	284	275	277	270	270	246	237
	Edh at Noon		21	27	32	33	34	35	36	31	28	21	18	18
All-Sky Solar Radiation	RadAvg		374	763	1162	1569	1674	1738	1774	1495	1075	618	324	253
	RadStd		46	71	97	126	102	136	140	130	101	86	34	31

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (9 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page